



12. What is the result of aggregating the following networks: 192.168.0.0/24, 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24?
- A) 192.168.0.0/22
 - B) 192.168.0.0/28
 - C) 192.168.0.0/30
 - D) 192.168.1.0/22
13. Which of the following is true about NAT?
- A) NAT can only be used in IPv6 networks.
 - B) NAT provides a way to conserve public IP addresses.
 - C) NAT always increases the available number of private IP addresses.
 - D) NAT prevents all unauthorized external traffic to a local network.
14. What is the primary purpose of tunneling in networking?
- A) To encrypt data before it is transmitted
 - B) To route data through a network using a private IP address
 - C) To reduce the bandwidth usage of a network
 - D) To allow one network protocol to be securely encapsulated within another network protocol
15. In the context of OpenFlow, what does the data plane refer to?
- A) The logical rules and policies applied by the controller
 - B) The set of devices that control the flow of data in the network
 - C) The hardware and software that process, forward, and drop network traffic
 - D) The protocols used to communicate between switches and routers
16. How would you configure an OpenFlow switch to block all datagrams destined to TCP port 22?
- A) The OpenFlow controller creates a flow entry with the destination port set to 22 and an action to forward all matching packets to the controller.
 - B) The OpenFlow controller creates a flow entry with the destination port set to 22 and an action to drop all matching packets.
 - C) The OpenFlow switch creates a flow entry for packets with source port 22 and drops them.
 - D) The OpenFlow controller configures the switch to forward packets to port 22 and ignores the destination port.
17. Which of the following is true about a MAC address?
- A) It is assigned dynamically by the router each time a device connects to the network.
 - B) It is a unique, hardware address burned into the network interface card (NIC) of the device.
 - C) It is used to identify the device's IP address across different networks.
 - D) It is only used by devices in the application layer to identify their respective users.
18. Which of the following is a common multiple access protocol used in the data link layer?
- A) Time Division Multiplexing (TDM)
 - B) Carrier Sense Multiple Access with Collision Detection (CSMA/CD)
 - C) Border Gateway Protocol (BGP)
 - D) Virtual Circuit Switching (VCS)
19. In ARP, what happens when a device sends out an ARP request?
- A) The device sends a broadcast message to the entire network asking, "Who has this IP address? Please send me your MAC address."
 - B) The device sends a unicast message to the specific device it wants to find the MAC address for.
 - C) The device sends an HTTP request to query the MAC address of the destination IP.
 - D) The device queries the network router to obtain the MAC address of the destination device.
20. Which of the following is true about VLANs?
- A) VLANs are only used to segregate networks by their physical locations.
 - B) A VLAN can be configured to group devices from different physical locations into the same broadcast domain.
 - C) VLANs can only be configured on Layer 3 (Network layer) devices like routers.
 - D) VLANs require a single physical switch to manage all the devices in the network.



1. In an IP datagram, what is the purpose of the Flags field?
- To indicate the version of the IP protocol used
 - To specify the source and destination IP addresses
 - To define the protocol (e.g., TCP, UDP) for transport layer communication
 - To control fragmentation and reassembly of packets
2. In the context of the IP header, the Fragment Offset field is used to:
- Indicate the total length of the fragmented datagram
 - Indicate the number of fragments created from the original datagram
 - Specify the position of the fragment in the original datagram
 - Define the fragmentation protocol being used
3. Which field in the IP datagram header helps to reassemble fragmented datagrams?
- Protocol
 - Source Address
 - Identification
 - Fragment Offset
4. Which of the following IPv4 addresses is a private IP address?
- 10.0.0.1
 - 172.20.5.4
 - 192.168.1.10
 - All of the above
5. What is the subnet mask for a network with a /30 prefix length?
- 255.255.255.0
 - 255.255.255.252
 - 255.255.255.240
 - 255.255.255.128
6. In the subnet 192.168.10.0/26, what is the broadcast address?
- 192.168.10.0
 - 192.168.10.63
 - 192.168.10.128
 - 192.168.10.127
7. What is the maximum number of hosts that can be accommodated in a /22 subnet?
- 1022
 - 510
 - 254
 - 128
8. In an IPv4 subnet, the network address and broadcast address cannot be assigned to hosts. Why?
- They are reserved for routers
 - They are used for routing purposes
 - They are reserved for internal use only
 - They are not valid IP addresses for devices
9. What is the primary function of the DHCP protocol?
- To encrypt data between devices on a network
 - To route data between different networks
 - To assign IP addresses dynamically to devices on a network
 - To assign static IP addresses to devices on a network
10. What does a DHCP Offer message contain?
- The client's current IP address
 - The IP address offered by the DHCP server, the lease duration, and other configuration details
 - The client's request for an IP address
 - Acknowledgement of the client's request for an IP address
11. If a client is not connected to the network when the DHCP lease expires, what will happen?
- The client will automatically renew the lease when it reconnects
 - The client will be assigned a new IP address
 - The DHCP server will release the IP address immediately
 - The client will keep its current IP address
12. When performing route aggregation, what does the common prefix represent?
- The subnet mask that applies to all IP addresses in the aggregated range
 - The first IP address in the aggregated block
 - The boundary point where addresses are split
 - The network address of the first address block

- congestion control algorithm is used by TCP to avoid congestion and control the data transmission rate?
 A) Slow Start B) Fast Retransmit C) Slow Start D) Adaptive Window Control
- Which of the following describes the "Additive Increase, Multiplicative Decrease" (AIMD) approach used in TCP's congestion control?
 A) TCP increases the congestion window linearly and reduces it by halving the size upon congestion detection
 B) TCP increases the congestion window size exponentially and reduces it by a fixed factor
 C) TCP uses a constant window size during congestion control
 D) TCP increases the congestion window exponentially and reduces it by a constant factor
- How does TCP detect congestion on a network?
 A) By measuring the transmission delay and round-trip time (RTT)
 B) By monitoring the available bandwidth on the link
 C) By receiving signals from network routers
 D) By checking for data packet acknowledgment and retransmissions
- In the slow-start phase, what happens when TCP receives three duplicate ACKs?
 A) It restarts the slow-start phase
 B) It reduces the congestion window by half
 C) It enters the fast recovery phase
 D) It moves into the congestion avoidance phase
- Which of the following is NOT a key responsibility of the network layer?
 A) Routing data packets from source to destination
 B) Providing error detection and correction
 C) Segmenting data into smaller packets for transmission
 D) Logical addressing of devices
4. What type of address is used by devices in the network layer to identify and communicate with other devices across different networks?
 A) MAC address B) IPv4 address C) URL address D) DNS address
5. Which of the following is NOT typically handled by the network layer?
 A) Path selection and routing B) Packet fragmentation and reassembly
 C) Error detection and correction D) Establishment of connections between devices
26. In a router, which scheduling mechanism is designed to provide fair allocation of bandwidth to all flows?
 A) Weighted Fair Queuing (WFQ) B) First-Come, First-Served (FCFS)
 C) Round-Robin Scheduling D) Priority Queuing (PQ)
27. Which field in the IP header specifies the total length of the datagram, including both header and data?
 A) Protocol B) Identification C) Time to Live (TTL) D) Total Length
28. The Time to Live (TTL) field in the IP header is used to:
 A) Indicate the priority of the packet
 B) Specify how long the packet is allowed to live in the network before being discarded
 C) Define the routing path for the packet
 D) Identify the source and destination IP addresses
29. The Protocol field in the IP header indicates:
 A) The type of the packet (e.g., TCP, UDP, ICMP)
 B) The total number of hops the packet has taken
 C) The destination port number
 D) The IP address of the next-hop router
30. The destination Address field in the IP header specifies:
 A) The type of the packet
 B) The IP address of the next-hop router
 C) The source IP address
 D) The MAC address of the sender

8) To control fragmentation and assembly of particles

1. The first of these is the fact that the
 2. of the system is not a simple matter.
 3. of the system is not a simple matter.
 4. of the system is not a simple matter.
 5. of the system is not a simple matter.
 6. of the system is not a simple matter.
 7. of the system is not a simple matter.
 8. of the system is not a simple matter.
 9. of the system is not a simple matter.
 10. of the system is not a simple matter.

一、（一） 凡有...
 二、（二） 凡有...
 三、（三） 凡有...
 四、（四） 凡有...
 五、（五） 凡有...
 六、（六） 凡有...
 七、（七） 凡有...
 八、（八） 凡有...
 九、（九） 凡有...
 十、（十） 凡有...

上ノ人ハ一ツ也
下ノ人ハ二ツ也
中ノ人ハ三ツ也
左ノ人ハ四ツ也
右ノ人ハ五ツ也
前ノ人ハ六ツ也
後ノ人ハ七ツ也
内ノ人ハ八ツ也
外ノ人ハ九ツ也
内ノ人ハ十ツ也

D) Define the fragmentation protocol being used.

Which of the following IPAT addresses is a loopback address?

A) 10.0.0.1
 B) 127.0.0.1
 C) 192.168.1.1
 D) 255.255.255.255

1. $\frac{1}{2} \log \frac{1}{2}$
 2. $\frac{1}{2} \log \frac{1}{2}$
 3. $\frac{1}{2} \log \frac{1}{2}$
 4. $\frac{1}{2} \log \frac{1}{2}$
 5. $\frac{1}{2} \log \frac{1}{2}$
 6. $\frac{1}{2} \log \frac{1}{2}$
 7. $\frac{1}{2} \log \frac{1}{2}$
 8. $\frac{1}{2} \log \frac{1}{2}$
 9. $\frac{1}{2} \log \frac{1}{2}$
 10. $\frac{1}{2} \log \frac{1}{2}$

1) 255.255.255.0
2) 255.255.255.128
3) 255.255.255.192
4) 255.255.255.224
5) 255.255.255.240
6) 255.255.255.248
7) 255.255.255.252
8) 255.255.255.254
9) 255.255.255.255
10) 255.255.255.256

the subnet 192.168.10.0/24

2. 168.10.0
B) 168.10.0.255

What is the maximum number of hosts that can be assigned to the 192.168.10.0/24 network?

C) 192.168.10.128
D) 192.168.10.127

8) 510
How many can be accommodated in a /22 subnet?

c) 254

They are reserved for routers
They are reserved for internal use only

is the primary function of the domain name system (DNS). It translates domain names into IP addresses. If they are not valid IP addresses for devices on the network, the DNS will return an error.

encrypt data between devices on a network
assign IP addresses to devices on a network

D) To assign static IP addresses to devices on a network

Does a DHCP Offer message contain?

client's current IP address

P address offered by the DHCP server, the lease duration, and other configuration details.

D) Acknowledgement of the client's request for an IP address

What happens if a client is not connected to the network when the DHCP lease expires, what will happen?

B) The client will be assigned a new IP address when it reconnects.

forming route aggregation, what does the router do to ensure that the client will keep its current IP address for

net mask that applies to all IP addresses in the aggregated range

boundary point where addresses are split:

U) The network address of the first address block



What is the maximum segment size (MSS) in the context of TCP?

- A) The maximum size of a data packet that a network can handle
- B) The maximum size of a segment that TCP can send without fragmentation
- C) The size of the TCP header
- D) The size of the data in the packet, excluding the TCP header

What is a common use case for UDP rather than TCP?

- A) Email transmission
- B) File transfer
- C) Real-time video and voice calls
- D) Web browsing

In the context of multiplexing, what is the role of port numbers at the transport layer?

- A) To identify the source and destination devices on a network
- B) To assign a unique identifier for each network in a multi-network system
- C) To distinguish between different application data streams on a single device
- D) To encode data for security purposes

Which of the following is TRUE regarding multiplexing in TCP?

- A) It uses port numbers to combine multiple data streams into a single stream
- B) It requires data to be encrypted before transmission
- C) It ensures that data is sent in real-time to applications
- D) It handles packet fragmentation during transmission

What part of a TCP or UDP segment is used to identify which application the data belongs to during demultiplexing?

- A) Sequence number
- B) Port number
- C) Checksum
- D) Source IP address

Which of the following is NOT a benefit of multiplexing in the transport layer?

- A) It reduces the number of network connections needed for multiple applications
- B) It allows multiple applications to share the same transport connection
- C) It guarantees the order of delivery of data for all applications
- D) It enables better utilization of the available bandwidth

Which of the following is NOT a feature of UDP?

- A) No connection establishment before data transfer
- B) Reliable data delivery with acknowledgments
- C) No flow control or congestion control
- D) Low overhead and faster transmission

How does UDP handle data transmission in terms of packet ordering?

- A) It ensures that packets arrive in the order they were sent
- B) It guarantees no loss of packets
- C) It does not guarantee packet order or delivery
- D) It sends data only after confirming receipt from the receiver

Which of the following is a key characteristic of UDP that makes it "stateless"?

- A) It establishes a connection before sending data
- B) It keeps track of data flow between sender and receiver
- C) It does not maintain any state or connection information between packets
- D) It retransmits lost data until it is acknowledged

In TCP, what is the role of the "ACK" (Acknowledgment) number?

- A) It identifies the sequence number of the next expected byte
- B) It indicates the number of bytes the receiver is willing to accept
- C) It is used to establish the connection between sender and receiver
- D) It provides information about the packet's status

What happens when the TCP sender's congestion window reaches the size of the receiver's window?

- A) The sender stops sending any further data
- B) The sender begins to reduce the transmission rate
- C) The sender continues sending, but waits for further acknowledgment
- D) The sender initiates a timeout to detect packet loss



Question 1: Multiple Choice Question

Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13	14
15	16	17	18	19	20	21	22	23	24	25	26	27	28
29	31	32	33	34	35	36	37	38	39	40	41	42	43
44	45	46	47	48	49	50	51	52	53	54	55	mark	

1. What is the main advantage of the User Datagram Protocol (UDP) over the Transmission Control Protocol (TCP)?
- A) UDP provides error checking and correction B) UDP guarantees reliable delivery of data
- C) UDP provides faster data transmission with minimal overhead D) UDP establishes a connection before sending

- Which TCP feature ensures that data is received in the correct order by the destination?
- A) Flow control B) Error detection C) Sequence numbers D) Acknowledgment

- Which of the following is true about the checksum used in TCP and UDP?
- A) It ensures that data is correctly delivered to the destination B) It is used only by the UDP protocol for error detection
- C) It prevents congestion in the network D) It checks for errors in the data during transmission

- The three-way handshake is a process used by which protocol?
- A) HTTP B) FTP C) TCP D) UDP

- What does the "window size" in TCP flow control indicate?
- A) The size of each segment in the TCP connection
- B) The amount of data that can be transmitted before an acknowledgment is required
- C) The maximum size of a single data packet in the connection
- D) The amount of memory allocated for connection management

- Which of the following is NOT a transport-layer service?
- A) Flow control B) Segmentation and reassembly C) Routing D) Error detection and correction

- What is the primary purpose of the checksum in both TCP and UDP?
- A) To establish the session between sender and receiver B) To provide encryption of data
- C) To prevent network congestion D) To detect errors in the transmission